2025 CSC493 – Capstone Weekly Reports[1]

<Download this template and name it Week_<WEEK_NUMBER>, where "<REPORT_NUMBER>" is replaced by the current week number (i.e., "1" for the first weekly report and so forth).

Week	6
Project Name	AI Music Mood Classifier
Report Date	Sep 30, 2025 ← click to select date

Part 1: Weekly Progress Report

- **Accomplishments**: **What did you complete this week?** Be specific and measurable.
- Examples: *Implemented login authentication using OAuth2; Fixed bug in file upload function (issues #14); Completed draft of UI wireframes; Identified functional and non-functional requirements; etc. (up to 3 points)*

This week, I added an SVM model and a RandomForest model to my baseline_classifier models list. I also added F1, ROC-AUC, confusion matrices, ROC curves to gauge accuracy of the models and also figure out what features each model is more reliant on using for gauging valence and arousal metrics. I also allowed for plotted graphs to be saved and also added the choice for graphs to not be displayed every run.

- **Challenges**: **What problems did you encounter?** Mention blockers, unresolved issues, or risks.
- Examples: *Struggled with connecting database migrations to production server; API documentation was incomplete – need clarification; etc. (up to 3 points)*

The challenge I have ran into this week and will probably run into for the rest of the weeks of this project is time management, as I am adding fairly simple additions to my project, but due to how many additions I am adding and the amount of time it takes to watch tutorials and implement them, it takes up a large portion of my time I spend on the project.

- **Next Steps**: **What are your goals for next week?** Keep them realistic and linked to project milestones.
- Examples: Set up automated testing with GitHub Actions; Final project proposal *(up to 2 points)*

Next week I plan to integrate my GUI skeleton with my functional models and build up the GUI and make sure it looks, works, and feels right.

- **Discussion Points**: **What would you like to discuss in class, or with your professor/TA?** This could include technical issues, project direction, or collaboration concerns.
- Examples: *Should we prioritize front-end features before database scaling? Need guidance on writing integration tests; etc. (up to 2 points)*

Since the GUI will eventually need to handle spectrograms for the CNN, should I design the interface now with that in mind, or just adapt it later once the CNN is working? How should I handle audio files that differ in length, quality, or format (e.g., mp3 vs. wav)? Should I enforce a standard format now or build in preprocessing for multiple types?

[1] Detailed Weekly Report requirements can be found here: 2024 URCPP Capstone - Using Agile and Reporting Out

Part 2: Time Reporting

Make sure that as you fill out the first prompt, you include in enough detail in the summary. For example, "debugging" is vague, but "debugged function X to make sure that when user does action Y, it is called and returns the value Z" is better.

- **Time Spent**: Briefly explain how much time, *outside of class*, spent on your project. If you worked on multiple components, each should get a detailed summary. Make sure to add up all the hours and minutes correctly. *Add as many rows as you need to the table below. Please do not include hours in class as part of this section. (up to 4 points)*

START	FINISH	HOURS	DETAILED SUMMARY
MM/DD HH:MM	MM/DD HH:MM		
9/23 08:00	9/23 12:00	4	Watched youtube tutorials of sklearn and how to implement SVM and RandomForest models, started adding those models
9/24 18:00	9/24 20:00	2	Worked more on adding the SVM and RandomForest models to the baseline classifiers
9/25 08:00	9/25 12:00	4	Finished adding SVM and RandomForest models to baseline classifiers.
9/26 16:00	9/26 18:00	2	Started adding F1, ROC-AUC, confusion matrices, ROC curves metrics as outputted plotted graphs for better gauging of accuracy per model and better understanding what features each model uses as more important inputs.
9/27 10:00	9/27 12:00	2	Finished adding F1, ROC-AUC, confusion matrices, ROC curves metrics.
	WEEKLY TOTAL	14	

- **Total (Cumulative) Project Time Spent**: After the number of hours and minutes, make sure to briefly explain whether you are on track and if not, what you may need to do in order to achieve what you set out to accomplish. *(up to 2 points)*

So far, I have spents 80 hours total on this project. I am on track according to the timeline built using the help of ChatGPT. During week 6, (September 30-October 5), I need to integrate the classifier into the GUI and get the GUI to be functional and look well.

Rubric:

The following rubric will be used, but they might change as needed.

Accomplishments (3 points)

1 point for a general description of progress, 2 points for specifics on progress, 3 points for specifics AND referring to previous targets and explaining how current accomplishments build on previous ones.

Challenges (3 points)

1 point for mentioning there are roadblocks, 2 points for specifics, 3 points for specifics AND what was done already to try to overcome them.

Desired discussion points (2 points)

1 point for at least one relevant discussion point as a general question, 2 points for relevant discussion points with specifics

Future Goals (2 points)

1 point for concrete future targets (i.e. "working more on the project" is a zero, but "working on getting component X to interface with component Y" suffices), 2 points for tying in the targets with what was hopefully discussed in the meeting.

Time Spent (4 points)

1 point for including general statements of how much time was spent ("4 hours on coding"), 2 points for splitting time into specific parts ("1.5 hours on research on component X, 1 hour coding, 2.5 hours debugging"), 3 points for specific parts and details on the pieces ("1.5 hours researching Turtle interface for drawing concentric circles given inputs from the user, 1 hour coding function X that used that interface, 2.5 hours testing function X by giving it multiple values and fixing errors for values A, B, C, and D"). 1 Point for totalling the hours correctly.

What happens if your time on a task is interrupted and you don't have a concrete (or discrete) end time? In this case put the start time in, and the word "interrupted" for the end time and include the task total time. Rounding to the nearest 15 minutes is acceptable. *(This makes adding up times easier, especially when you use decimal hours, i.e. 3.75 rather than 3 hours 45 minutes.)*

Total (Cumulative) Project Time (2 points)

1 point for summing the values correctly, 2 points for the total time AND reflection on progress (you are confident to fit the target and if not, what course corrections you anticipate needing to make)

Resources

Here's a link to this Weekly Report Template – Make a copy and use it:

2025 CSC493 Weekly Report v2 TEMPLATE